



Cyberscope

A **TAC Security** Company

Audit Report

Bigtr

July 2026

File name BigTrPresale.sol

SHA256 d45e7a636ac04fa64744556724d54a548e27b683a743fa917bad2c4fcea5fed

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Risk Classification

The criticality of findings in Cyberscope's smart contract audits is determined by evaluating multiple variables. The two primary variables are:

1. **Likelihood of Exploitation:** This considers how easily an attack can be executed, including the economic feasibility for an attacker.
2. **Impact of Exploitation:** This assesses the potential consequences of an attack, particularly in terms of the loss of funds or disruption to the contract's functionality.

Based on these variables, findings are categorized into the following severity levels:

1. **Critical:** Indicates a vulnerability that is both highly likely to be exploited and can result in significant fund loss or severe disruption. Immediate action is required to address these issues.
2. **Medium:** Refers to vulnerabilities that are either less likely to be exploited or would have a moderate impact if exploited. These issues should be addressed in due course to ensure overall contract security.
3. **Minor:** Involves vulnerabilities that are unlikely to be exploited and would have a minor impact. These findings should still be considered for resolution to maintain best practices in security.
4. **Informative:** Points out potential improvements or informational notes that do not pose an immediate risk. Addressing these can enhance the overall quality and robustness of the contract.

Severity	Likelihood / Impact of Exploitation
● Critical	Highly Likely / High Impact
● Medium	Less Likely / High Impact or Highly Likely/ Lower Impact
● Minor / Informative	Unlikely / Low to no Impact

Review

Audit Updates

Initial Audit	16 Jul 2026 https://github.com/cyberscope-io/audits/blob/main/bigtr/v1/audit.pdf
Corrected Phase 2	23 Jul 2026

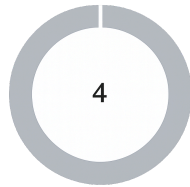
Source Files

Filename	SHA256
BigTrPresale.sol	d45e7a636ac04fa64744556724d54a548e27b683a743fa917bad2c4fcea e5fed

Overview

BigTrPresale is a single contract that runs a capped stage token presale. Buyers pay one whitelisted 18 decimal stablecoin to buy BIGTR allocations, priced on a stage schedule that steps up as each stage fills, with cross stage purchases split and priced correctly in one transaction, and buyers can set a floor on how many tokens they are willing to accept for a given payment. Payment goes straight to a treasury, and the contract only records each buyer's allocation for a separate distribution contract to pay out later it never holds or delivers BIGTR itself.

Findings Breakdown



● Critical	0
● Medium	0
● Minor / Informative	4

Severity	Unresolved	Acknowledged	Resolved	Other
● Critical	0	0	0	0
● Medium	0	0	0	0
● Minor / Informative	4	0	0	0

Diagnostics

● Critical ● Medium ● Minor / Informative

Severity	Code	Description	Status
●	CDCO	Ceiling Division Can Overflow	Unresolved
●	CCR	Contract Centralization Risk	Unresolved
●	DCT	Distribution Contract Trust	Unresolved
●	PRA	Payment Routing Assumption	Unresolved

CDCO - Ceiling Division Can Overflow

Criticality	Minor / Informative
Location	BigTrPresale.sol#L258,288,305
Status	Unresolved

Description

The `_ceilDiv` helper computes ceiling division as $(a + b - 1) / b$. When `a` is close enough to `type(uint256).max`, the intermediate addition can overflow even though the final quotient would fit within `uint256`. Solidity's checked arithmetic consequently reverts the operation.

The helper is used for stage-cost calculations in `_simulate`, so affected stage parameters can cause `quote` and `buy` to revert, potentially making the corresponding inventory unsellable. For nonzero `a`, ceiling division can instead be expressed using $((a - 1) / b) + 1$, which avoids the overflowing addition.

```
function _ceilDiv(uint256 a, uint256 b) internal pure returns (uint256)
{
    return (a + b - 1) / b;
}
```

Recommendation

Consider replacing the addition-based calculation with an overflow-safe ceiling-division method that divides after subtracting from the numerator. Explicit handling for zero numerators and divisors should be retained, together with boundary tests covering values near `type(uint256).max`.

CCR - Contract Centralization Risk

Criticality	Minor / Informative
Location	BigTrPresale.sol#L118
Status	Unresolved

Description

The contract's functionality and behavior are heavily dependent on external parameters or configurations. While external configuration can offer flexibility, it also poses several centralization risks that warrant attention. Centralization risks arising from the dependence on external configuration include Single Point of Control, Vulnerability to Attacks, Operational Delays, Trust Dependencies, and Decentralization Erosion.

```
function setPayToken(address token, bool accepted) external onlyOwner
function setTreasury(address t) external onlyOwner
function pause() external onlyOwner { _pause(); }
function unpause() external onlyOwner { _unpause(); }
function finalize() external onlyOwner
```

Recommendation

To address this finding and mitigate centralization risks, it is recommended to evaluate the feasibility of migrating critical configurations and functionality into the contract's codebase itself. This approach would reduce external dependencies and enhance the contract's self-sufficiency. It is essential to carefully weigh the trade-offs between external configuration flexibility and the risks associated with centralization.

DCT - Distribution Contract Trust

Criticality	Minor / Informative
Location	BigTrPresale.sol#L70
Status	Unresolved

Description

This contract never hands out real `BIGTR` tokens. It only writes down how many tokens each buyer earned, in the `tokensAllocated` mapping. A separate contract, not part of this review, is meant to read that number later and pay it out. Two things depend entirely on that other contract being right it needs to use the same 18 decimals this contract assumes, and it needs to stop anyone from claiming their allocation twice. We can't check either one without seeing that code.

```
mapping(address => uint256) public tokensAllocated; // cüzdanın hak ettiği BIGTR (wei)
```

Recommendation

It is recommended that the team review the distribution contract with the same care once it's written, and double check that its decimal count and claim logic match what this contract assumes.

PRA - Payment Routing Assumption

Criticality	Minor / Informative
Location	BigTrPresale.sol#L18,232
Status	Unresolved

Description

The comments at the top of this file say that payments made in BNB, ETH, Polygon, or by card get turned into stablecoin by a frontend service before they reach `buy()`. That means `buy()` might sometimes get called by that service on a buyer's behalf, rather than by the buyer's own wallet directly. `tokensAllocated` only ever credits whoever calls `buy()`, whatever address that turns out to be. If that outside service ever calls `buy()` using a shared address, or a short lived one, instead of an address the buyer truly owns, that buyer's allocation could end up recorded somewhere they can't reach later.

```
tokensAllocated[msg.sender] += tokens;
```

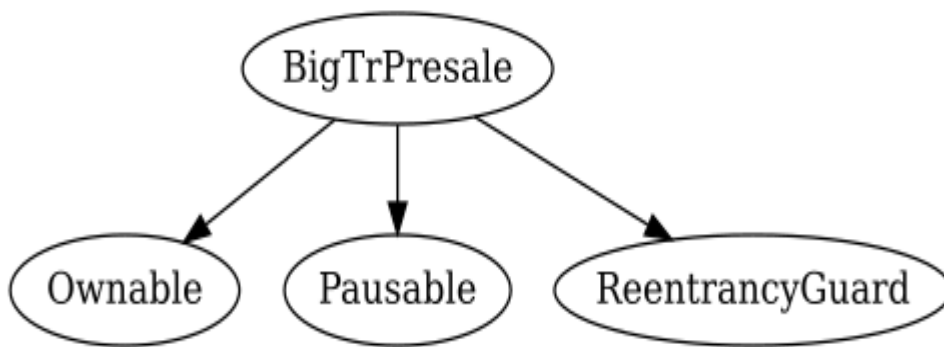
Recommendation

The team is advised to confirm exactly which address calls `buy()` for every payment path, especially the ones that don't pay in stablecoin directly, and make sure that address always belongs to the actual buyer, not a shared or temporary one.

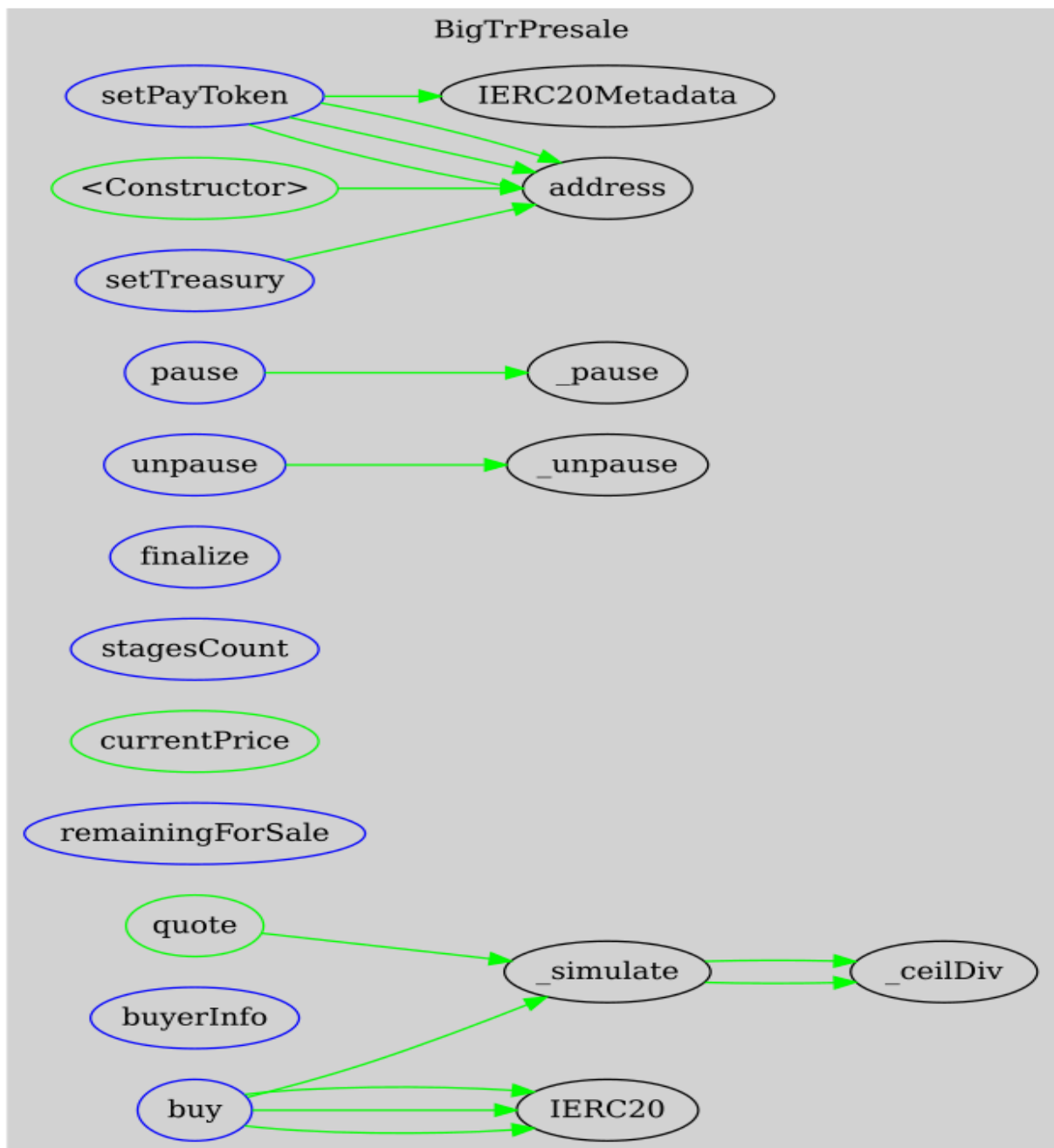
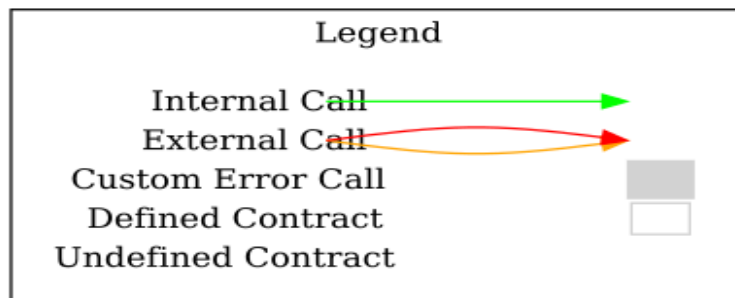
Functions Analysis

Contract	Type	Bases		
	Function Name	Visibility	Mutability	Modifiers
BigTrPresale	Implementation	Ownable2Step, Pausable, ReentrancyGuard		
		Public	✓	Ownable
	setPayToken	External	✓	onlyOwner
	setTreasury	External	✓	onlyOwner
	pause	External	✓	onlyOwner
	unpause	External	✓	onlyOwner
	finalize	External	✓	onlyOwner
	stagesCount	External		-
	currentPrice	Public		-
	remainingForSale	External		-
	quote	Public		-
	buyerInfo	External		-
	buy	External	✓	nonReentrant whenNotPaused
	_ceilDiv	Internal		
	_simulate	Internal		

Inheritance Graph



Flow Graph



Summary

BigTrPresale implements a capped, stage based token presale built around automatic price progression, same transaction cross stage purchase splitting, a single active payment token, and on chain recording of buyer allocations for later distribution, funded through direct stablecoin payment to a treasury on every purchase. This audit investigates security issues, business logic concerns and potential improvements.

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About Cyberscope

Cyberscope is a TAC blockchain cybersecurity company that was founded with the vision to make web3.0 a safer place for investors and developers. Since its launch, it has worked with thousands of projects and is estimated to have secured tens of millions of investors' funds.

Cyberscope is one of the leading smart contract audit firms in the crypto space and has built a high-profile network of clients and partners.



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The Cyberscope team

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